Inverse & Composite Functions

ANSWER KEY

Section 1: Composition of Functions and Domain

1 Given $f(x) = 2x + 5$ and $g(x) = x^2 - 1$, find $(f \circ g)(x)$.

a. $2x^2 + 3$ ✓

b. $4x^2 + 20x + 24$

c. $2x^2 + 9$

d. $2x^3 - 1$

$(f \circ g)(x) = f(g(x)) = 2(x^2 - 1) + 5 = 2x^2 - 2 + 5 = 2x^2 + 3$.

2 Given $f(x) = \sqrt{x - 2}$ and $g(x) = 3x + 1$, what is the domain of $(f \circ g)(x)$?

a. $\left[\frac{1}{3}, \infty\right)$ ✓

b. $[2, \infty)$

c. $\left(-\infty, \frac{1}{3}\right]$

d. $(-\infty, \infty)$

$(f \circ g)(x) = \sqrt{3x + 1 - 2} = \sqrt{3x - 1}$. For the square root to be real, $3x - 1 \geq 0 \Leftrightarrow 3x \geq 1 \Leftrightarrow x \geq \frac{1}{3}$.

3 If $h(x) = (3x - 4)^5$, which two functions f and g satisfy $h(x) = (f \circ g)(x)$?

a. $f(x) = x^5$ **and** $g(x) = 3x - 4$ ✓

b. $f(x) = 3x - 4$ and $g(x) = x^5$

c. $f(x) = 3x^5$ and $g(x) = x - 4$

d. $f(x) = (x - 4)^5$ and $g(x) = 3x$

Let inner function $g(x) = 3x - 4$ and outer function $f(u) = u^5$. Then $f(g(x)) = (3x - 4)^5$.

Section 2: Finding Inverse Functions

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4 Find the inverse function of $f(x) = \frac{3x-1}{2x+5}$.

a. $f^{-1}(x) = \frac{5x+1}{3-2x}$ ✓

b. $f^{-1}(x) = \frac{2x+5}{3x-1}$

c. $f^{-1}(x) = \frac{5x-1}{2x-3}$

d. $f^{-1}(x) = \frac{3-2x}{5x+1}$

Set $y = \frac{3x-1}{2x+5} \Rightarrow y(2x+5) = 3x-1 \Rightarrow 2xy+5y = 3x-1 \Rightarrow 5y+1 = x(3-2y) \Rightarrow x = \frac{5y+1}{3-2y}$. Swap variables: $f^{-1}(x) = \frac{5x+1}{3-2x}$.

5 What is the inverse function of $g(x) = x^3 + 7$?

a. $g^{-1}(x) = \sqrt[3]{x-7}$ ✓

b. $g^{-1}(x) = \sqrt[3]{x} - 7$

c. $g^{-1}(x) = (x-7)^3$

d. $g^{-1}(x) = \frac{1}{x^3+7}$

$y = x^3 + 7 \Rightarrow y - 7 = x^3 \Rightarrow x = \sqrt[3]{y-7}$. Swap variables: $g^{-1}(x) = \sqrt[3]{x-7}$.

6 If $f(x) = (x-3)^2$ for $x \geq 3$, what is $f^{-1}(x)$ and its domain?

a. $f^{-1}(x) = \sqrt{x} + 3$ **for** $x \geq 0$ ✓

b. $f^{-1}(x) = \sqrt{x+3}$ for $x \geq -3$

c. $f^{-1}(x) = -\sqrt{x} + 3$ for $x \geq 0$

d. $f^{-1}(x) = \sqrt{x} - 3$ for $x \geq 0$

Set $y = (x-3)^2$. Since $x \geq 3$, $x - 3 \geq 0$, so $\sqrt{y} = x - 3 \Rightarrow x = \sqrt{y} + 3$. Since the range of f is $[0, \infty)$, the domain of f^{-1} is $x \geq 0$.

Section 3: Graphical Symmetry & Invertibility

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7 The graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of each other across which line?

a. $y = x$ ✓

b. $y = -x$

c. $x = 0$ (y-axis)

d. $y = 0$ (x-axis)

Because the inverse interchanges the inputs and outputs $(a, b) \leftrightarrow (b, a)$, the graph of f^{-1} is the reflection of the graph of f across the line $y = x$.

8 Which test determines if a function f has an inverse that is also a function?

a. **Horizontal Line Test** ✓

b. Vertical Line Test

c. First Derivative Test

d. Ratio Test

A function is one-to-one (and thus has an inverse function) if and only if no horizontal line intersects its graph more than once (Horizontal Line Test).

9 If $f(2) = 7$ and $f'(2) = 4$, what is $(f^{-1})(7)$?

a. 2 ✓

b. 7

c. $\frac{1}{4}$

d. $\frac{1}{2}$

By definition of the inverse function, $f(a) = b \Leftrightarrow f^{-1}(b) = a$. Since $f(2) = 7$, $f^{-1}(7) = 2$.

Section 4: Verifying Inverses Algebraically

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10 Which condition algebraically proves that $f(x)$ and $g(x)$ are inverse functions?

a. $f(g(x)) = x$ **for all** x **in the domain of** g**, and** $g(f(x)) = x$ **for all** x **in the domain of** f ✓

b. $f(x) \cdot g(x) = 1$

c. $f(x) + g(x) = 0$

d. $f(g(x)) = g(f(x)) = 1$

Two functions f and g are inverses if and only if both compositions simplify to the identity function: $f(g(x)) = x$ and $g(f(x)) = x$.

11 Let $f(x) = 5x - 9$ and $g(x) = \frac{x+9}{5}$. Evaluate $f(g(x))$.

a. x ✓

b. $5x$

c. $x - 9$

d. 1

$f(g(x)) = 5\left(\frac{x+9}{5}\right) - 9 = (x + 9) - 9 = x$.

12 If a one-to-one function f has domain $[2, 10]$ and range $[-3, 5]$, what are the domain and range of f^{-1}?

a. **Domain:** $[-3, 5]$**; Range:** $[2, 10]$ ✓

b. Domain: $[2, 10]$; Range: $[-3, 5]$

c. Domain: $[-5, 3]$; Range: $[-10, -2]$

d. Domain: $[1/2, 1/10]$; Range: $[-1/3, 1/5]$

For an inverse function, the domain of f^{-1} equals the range of f ($[-3, 5]$), and the range of f^{-1} equals the domain of f ($[2, 10]$).

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